

Predictions: Theory and Practice

Part II: Subjective judgement

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Heuristics and Biases

Introduction

- Terminology
 - Heuristics: Mental shortcuts.
 - Biases: Systematic deviations from rationality.
- Evolutionary perspective
 - human perception and decision process adapted to certain environment
 - heuristics for forming beliefs developed
 - but those heuristics can be wrong for other environments
- nature vs nurture?
 - always hard to tell
 - but also culture subject to survival of the fittest
 - biases remain relevant irrespective of origin

Bias

- belief biases as short-hand for “errors in probabilistic reasoning.”
 - deviation in probabilities
 - or from Bayesian updating
- two possible origins
 - imperfect compliance to probability theory
 - mental model that is not accurate for reality
- identifying heuristics and biases from experiments and examples is difficult
 - many of the presented conclusions are contested or not universal
 - you should be aware that the biases might be relevant

Goals

- learn common heuristics in belief formation
- learn about biases arising from them

Literature

- partly based on Benjamin (2019)
- original research program by Kahnemann and Tversky
 - see Kahneman et al. (1982)
 - set of experiments that reveals common patterns in belief formation and decision making
- behavioral economics more focused on irrational decisions
 - irrational decisions follow from irrational predictions

Biases on samples

Partition dependence

- $P(A) < P(A \cap B) + P(A \cap B^c)$
- people assign greater total probability to an event when it is described as the union of subevents

Examples:

- What is the probability that the next person coming through the door is
 - ... a lawyer?
 - ... a lawyer and wears a suit?
- What is your probability of a
 - i. cardiovascular disease
 - ii. heart attack
 - iii. stroke

Sample-size neglect

- Law of large numbers: For $y_i \sim_{iid} B(p)$, it holds that

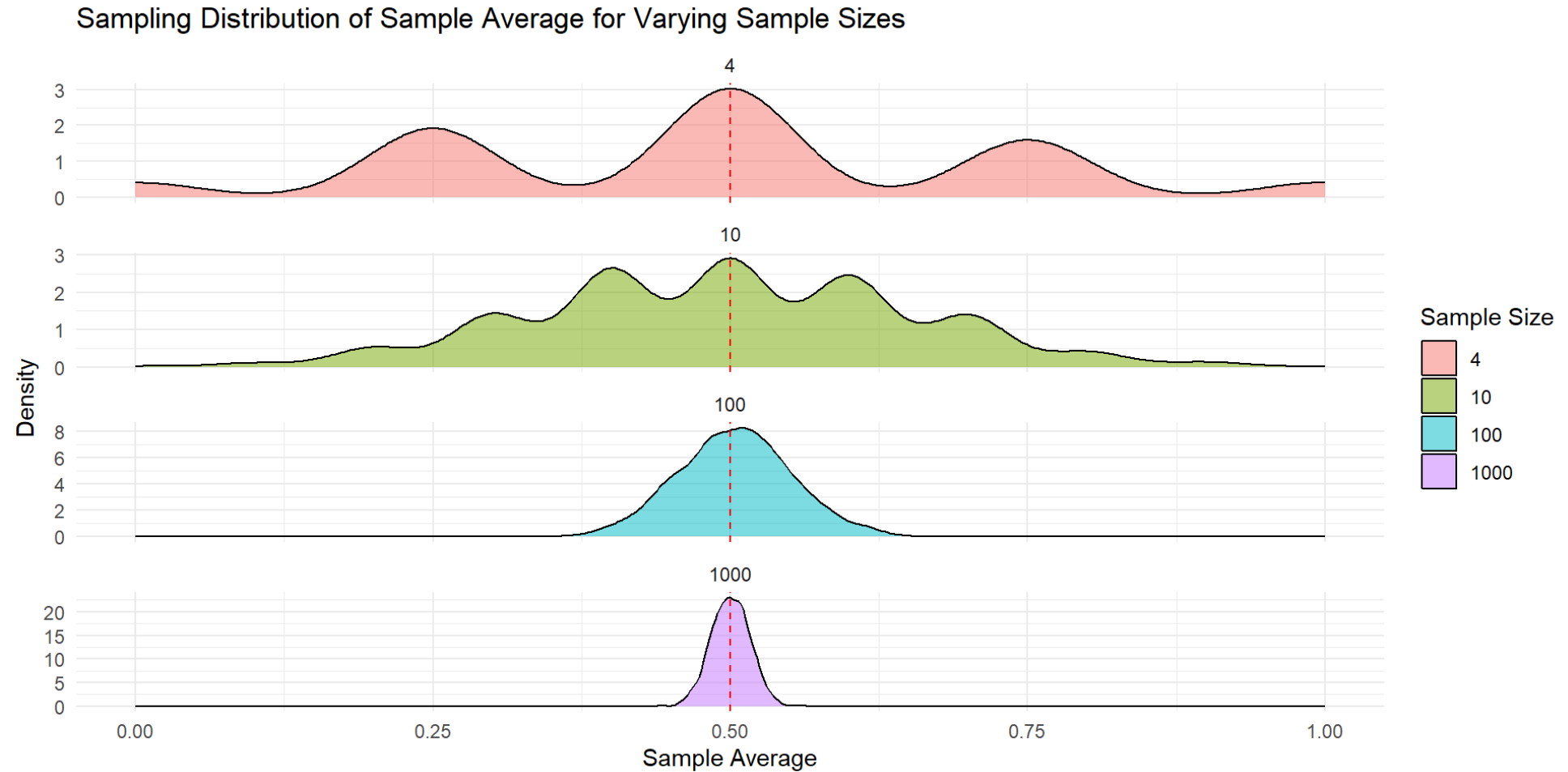
$$\frac{1}{n} \sum y_i \rightarrow_p p$$

- puts more mass around the probability, the larger the sample size
- **sample-size neglect:** sample size not sufficiently accounted for

Example:

- What is the distribution of the ratio of girls among 4 babies?
 - ...among 10 babies?
 - ...among 100 babies?

Simulation: True distribution of sample average



Overconfidence

- Overestimating the accuracy of predictions
- also called overprecision
 - overconfidence used more generally for overestimating own performance (also beyond predictions)
- for real-valued targets, the prediction is too sharp
 - not measurable with point forecasts alone

Example: 90% Confidence Interval

- Ask: give a range you are 90% sure contains the true answer
 - e.g., “In what year was the Eiffel Tower built?” → [lower, upper]
 - if you are well-calibrated, true values fall inside your range 90% of the time
- Typical result: people are right only **40–60%** of the time
 - intervals are far too narrow

Ignoring non-randomness

- random samples are commonly the basis in statistics
- but non-random samples are everywhere
 - running a race / driving on the street
 - number of friends
 - communication on positive / negative aspects
 - number of students in class
 - time series data

Biases on sequences

Gambler's fallacy

- “Observing one event reduces the likelihood of observing it again”
- wrong for independent samples
- especially wrong for independent identically distributed samples, where p is unknown
 - after observing the event, you update p to a higher posterior value
- Examples:
 - In gambling: “I lost the last rounds, now I am more likely to win.”
 - coin tosses: “After observing four times head, the next toss is more likely tails.”
 - “They had three boys in a row. The fourth child will probably be a girl.”

Hot Hand Fallacy

- “A streak of successes increases the probability of the next success”
- Opposite error to the gambler’s fallacy
- Original example: basketball
 - after several consecutive made shots, the next shot “feels” more likely to go in
- Also appears in: investment returns, creative output, sports performance

Law of Small Numbers (LSN)

- “a small sample should look representative of the population”
 - people expect even short sequences to mirror the long-run frequency
- Example: a fair coin flipped 6 times
 - sequence H-H-H-T-T-T feels *less* random than H-T-H-T-H-T, even though both are equally likely
 - sequence H-H-H-H-H-H feels *suspicious*, as if a correction is due
- **Connection to gambler’s fallacy:** LSN is the root cause
 - if you believe small samples must be representative, then after too many heads you expect the sequence to “self-correct”

Biases in updating

- consider Bayes updating with states A and B of the world and a signal S

$$P(A|S) = \frac{P(S \& A)}{P(S)}$$

- Or in its **posterior-odds** form

$$\frac{P(A|S)}{P(B|S)} = \frac{P(S \& A)}{P(S \& B)} = \frac{P(S|A)}{P(S|B)} \cdot \frac{P(A)}{P(B)}$$

- $P(S|A)$ is the likelihood of signal S given state A.
- $P(A)$ is the prior probability of state A.

Experimental setting

- a typical experiment would look like this
 - two urns named A and B with red and blue balls
 - urn A: 90% red and 10% blue
 - urn B: 10% red and 90% blue
 - unknown which type of urn this is
 - signal S = the result of some draws from the urn (balls are returned after each draw)
- What do we learn from a signal $S = RRBR$?
 - more red, more likely to be urn A
 - what is your probability prediction for it being urn A?
- Bayes solution
 - $p(A), p(B) = 0.5$ (if urn randomly chosen)
 - $p(S = RRBR|A) = 0.9^3 \times 0.1^1$

Experiment Analysis

- Bayes solution can be computed for

$$\frac{P(A|S)}{P(B|S)} = \underbrace{\frac{P(S|A)}{P(S|B)}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{P(A)}{P(B)}}_{\text{prior ratio}}$$

- but hard to do in your head (participants normally don't do it)
- How can we analyse deviations from Bayesian updating?
 - take log of equation to linearize

$$\log\left(\frac{P(A|S)}{P(B|S)}\right) = \log\left(\frac{P(S|A)}{P(S|B)}\right) + \log\left(\frac{P(A)}{P(B)}\right)$$

- posterior ratio we measure
- likelihood ratio we can compute
- prior ratio we can experimentally control

Empirical findings

- If we analyze the equation

$$\log\left(\frac{P(A|S)}{P(B|S)}\right) = \log\left(\frac{P(S|A)}{P(S|B)}\right) + \log\left(\frac{P(A)}{P(B)}\right)$$

in experiments we find:

- underinference is common (effect of likelihood ratio is smaller)
- underinference is larger if sample size is large
 - participants learn not enough from observing one signal
 - but discrepancy is even worse for multiple signals
- base rate neglect is common (effect of prior ratio is smaller)

Anchoring

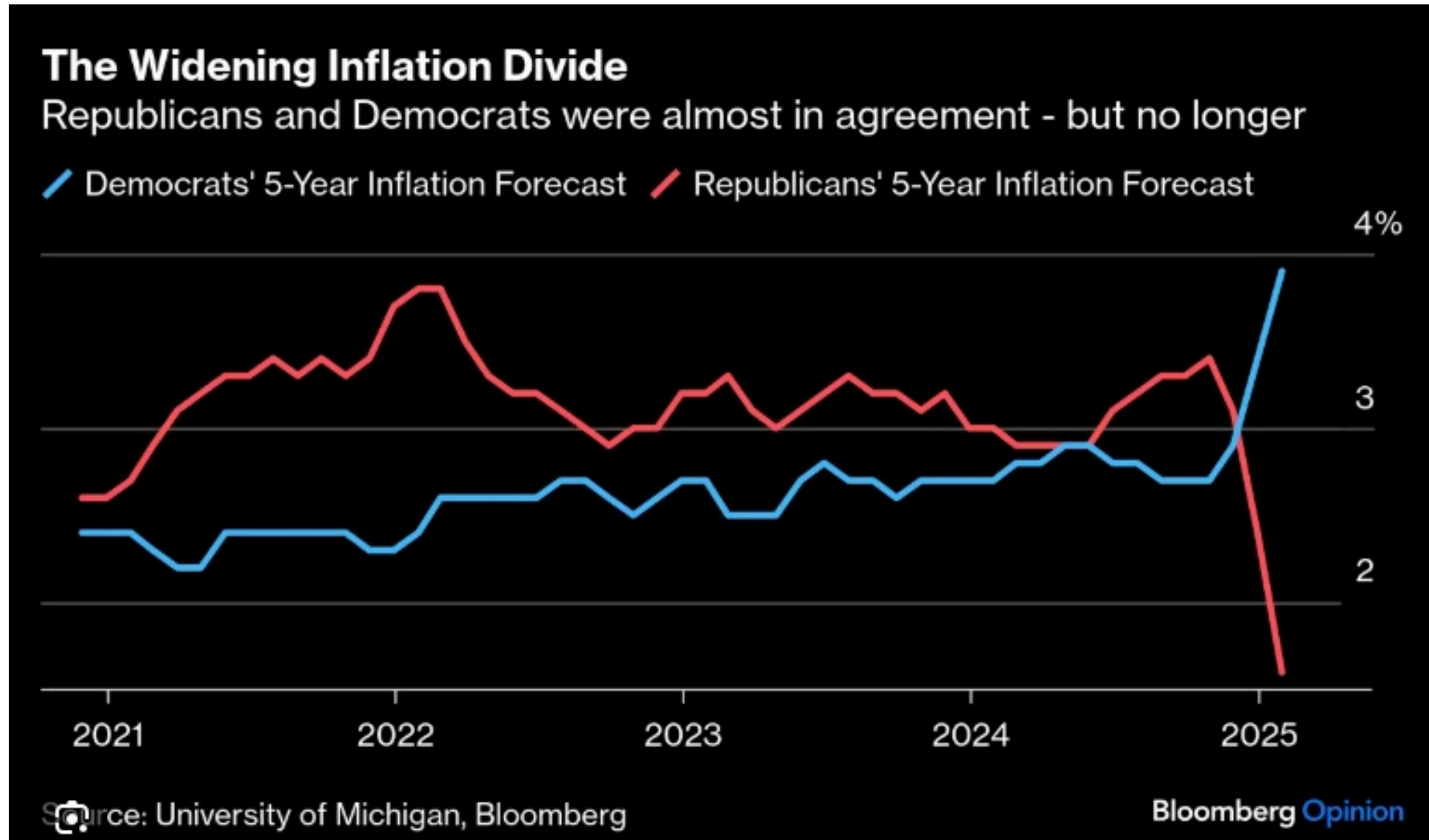
- Initial information (anchor) influences judgments
- even when the anchor is irrelevant
- Example:
 - Write the last two digits of your social security number down
 - What would you pay for getting this chocolat bar?
 - Result: Offers more than 60% higher if social security number is high

Motivated beliefs

The standard model postulates a separation of preferences (utility function) and beliefs (probabilities)

- but empirically this separation is consistently violated
- holding a belief can provide direct utility
 - ego protection
 - identity
 - wishful thinking

Example: Partisan inflation expectations



Example: Scams

- online scams often start with small money amounts and escalate later
- why? Let's think about the probability that a request is a scam.
 - unknown persons asks for money $\Rightarrow$ probably scam
 - unknown persons asks for more money after I already gave some $\Rightarrow$ even higher probability of scam, but acknowledgement would hurt.
-

Confirmation Bias

- Confirmation bias is the tendency to search for, interpret, focus on and remember information in a way that confirms one's preconceptions.
- Seeking/interpreting confirming information.
- Ignoring contradictory evidence.
- “Echo chambers”
- Affects information search and interpretation, leading to biased (and overprecise) predictions.

Hindsight Bias

- “I knew it all along”: after learning an outcome, people believe they had predicted it all along
 - memory of prior beliefs is distorted toward the observed outcome
- Undermines learning from experience
 - if you remember predicting correctly, you don’t update your method

Heuristics

Availability Heuristic

- Ease of recall influences probability judgments ($\hat{p}$)
 - recall more available, higher prediction $\hat{p}$
 - e.g., vividness distorts predictions of rare but salient events such that $E[y|\hat{p}] < \hat{p}$
- Examples:
 - Plane crashes (vivid)
 - car accidents (less vivid, but more dangerous)
 - terror attacks (vivid)
 - obesity (less vivid, but more casualties)

Example: 9/11 terrorist attacks made risk of flying more salient

“In the aftermath of the terrorist attacks on September 11 (9/11), 2001, from October 2001 to September 2002, there were approximately 1,600 more traffic fatalities in the United States than would normally be expected.” (Gaissmaier and Gigerenzer 2012)

- Idea:
 - perceived risk of flying increases
 - people fly less, some drive instead
 - driving is risky (in fact, 30 minutes taxi ride to airport more risk than a 5 hours flight)
 - indirect effect: more traffic accidents

Representativeness Heuristic

- Judging probability based on similarity to a prototype.
 - entails Base-rate neglect.
- Example: a person who is dressed in eccentric clothes and reading a poetry book.
 - poet or accountant?
 - looks like a typical poet
 - but many more people are accountants than poets

Biases in decisions

- Ambiguity aversion and probability weighting originate in decision theory
- connection between reported probability prediction and decisions often not investigated in literature
- human participants act as if they have these predictions

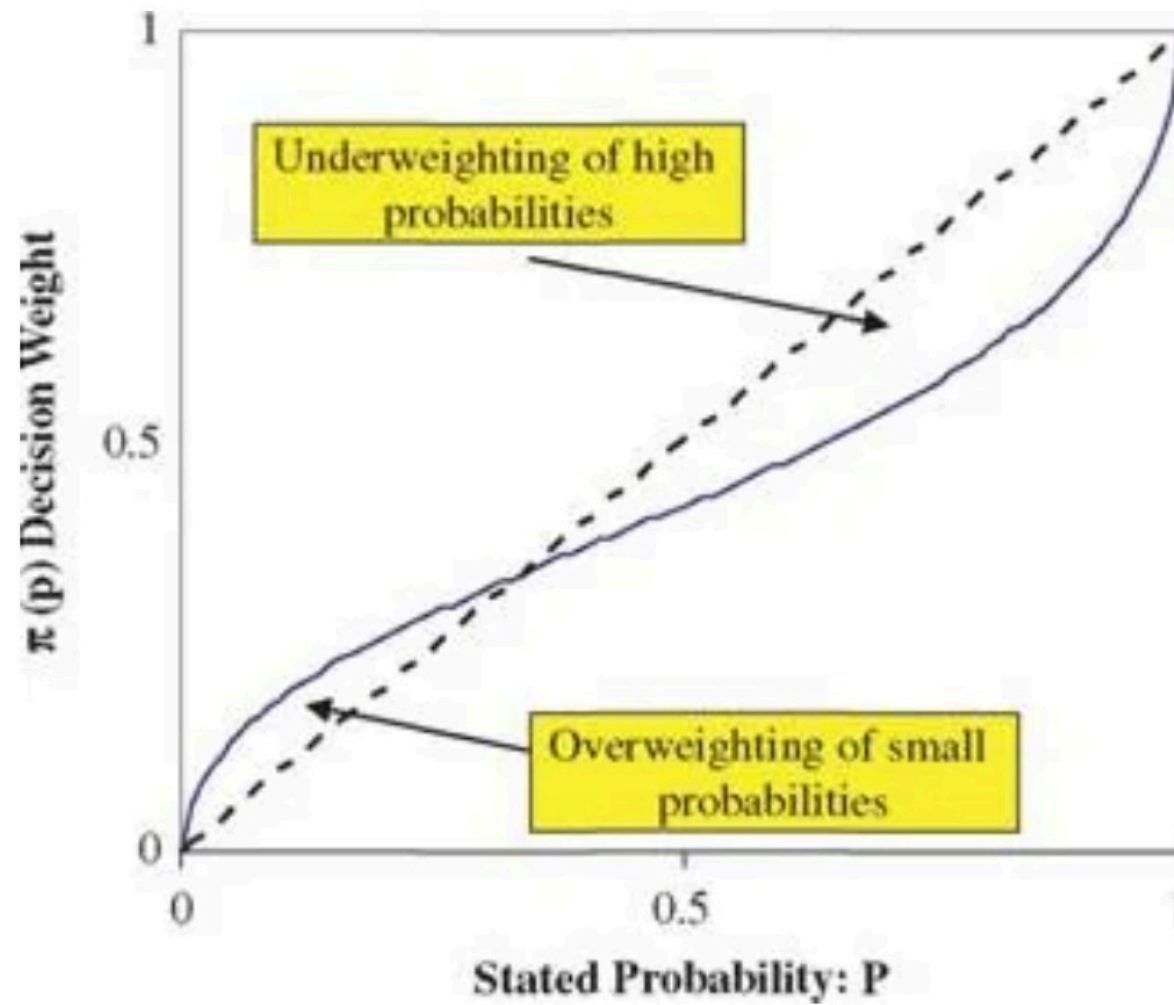
Ambiguity Aversion

- Preference for known risks over unknown risks.
- Affects mostly decisions in high uncertainty
- Not necessarily a bias
 - can be understood as more general decision-theory
- Connection to prediction performance unclear

Probability Weighting Function

- Distorting probabilities, especially small and large.
- Inverse S-shape.
- Small probabilities overweighted.
- Large probabilities underweighted.
- discrimination between probabilities imperfect
- really strong probability weighting:
 - differentiate only between yes, no, maybe
 - no difference between $p = 0.3$ and $p = 0.7$

Prob weighting function



From Problems to Solutions

- We have seen many systematic ways human judgment fails:
 - biases on samples and sequences (sample-size neglect, overconfidence, gambler's fallacy, hot hand fallacy)
 - biases in updating (base rate neglect, anchoring, motivated beliefs, confirmation bias, hindsight bias)
 - biases in decisions (ambiguity aversion, probability weighting)
 - heuristics that distort probability judgments (availability, representativeness, conjunction fallacy)
- **Key question:** Are these failures inevitable, or can they be overcome?
- The superforecaster research program asks: what do the *best* human forecasters do differently?

Superforecasters

Introduction

- What characterizes good human judgement?
- based on the research program by Tetlock and coauthors ([Mellers et al. 2015](#); [Tetlock and Gardner 2016](#))
 - initially funded by the U.S. intelligence community.
 - “superforecasters”: ordinary people who consistently outperformed intelligence analysts with access to classified information.
- focused almost exclusively on event predictions
 - performance criterium: Quadratic score (Brier score)

You can participate, too!

- popular options
 - good judgement open (prediction polls)
 - Metaculus (prediction poll)
 - PredictIt (Prediction market)
- no big European alternative
- our Assignment is essentially a forecasting competition
 - i. define targets
 - ii. predict

why human judgement?

- if there is data available, use statistical algorithms
- human judgement is used in situations, when there is no good data
- We will see with Bayesian inference: Difference vanishes to some point
 - every model can be traced back to assumptions which are ultimately based on human judgement
 - every expert can use data and algorithms to inform their judgement

The Parable: The Hedgehog and the Fox

“The fox knows many things, but the hedgehog knows one big thing.”

- the Hedgehog
 - **Knows “One Big Thing”:**
 - Relates everything to it
- the Fox
 - **Knows “Many Little Things”:**
 - Draws on a wide variety of experiences and information
 - Tolerant of conflicting perspectives

Hedgehog vs. Fox: Forecasting Accuracy

- Tetlock (**Tetlock and Gardner 2016**):
 - **Foxes consistently outperformed hedgehogs** in making accurate forecasts
 - Hedgehogs, particularly famous ones often sought by media, tended to be *less* accurate, sometimes worse than chance.
 - Overconfidence was a key factor in hedgehog inaccuracy.

Ingredients to super forecasting

- Superforecasters cultivate specific habits of mind and work methods
- These skills can be learned and improved
- Let's look at the key characteristics...

Property: Reasoning - Numerate & Probabilistic

- **Definition:** Using numbers and probabilities to express degrees of belief
- **Why it Matters:**
 - Avoids the ambiguity of vague words (“maybe,” “likely,” “possible”).
- **How to Improve:**
 - Practice translating vague feelings into specific percentages
 - Learn basic probability concepts and Bayes’ Theorem (even conceptually)
 - track your probabilistic forecasts vs. outcomes

Example: Finding Osama bin Laden

- search for Osama bin Laden
 - early 2011 the intelligence community was growing more confident
 - eventually communicated to President Obama
 - presentations about the likelihood that Osama was indeed residing in compound in a Pakistani town
- Multiple experts with probability statement between 35% to 95%
 - with a median of 75%
- President Obama:
 - “Seems like it’s a coin flip, 50-50.”

Property: Cautious Humility

“Beliefs are hypotheses to be tested, not treasures to be protected.”

- **Definition:** Acknowledging uncertainty, the limits of one’s own knowledge, and the role of chance. Understanding the world is complex and often unpredictable.
- **Why it Matters:**
 - more receptive to new evidence and correction
 - reduces overconfidence (overprecision) bias

Property: Thinking Style - Actively Open-Minded & Curious

- **Definition:** *Actively* seeking out perspectives and evidence that challenge one's own beliefs.
- **Why it Matters:**
 - Counteracts confirmation bias and groupthink
 - Leads to more comprehensive synthesis of information
- **How to Improve:**
 - Deliberately seek out sources and arguments that contradict your viewpoint
 - try to understand their reasoning
 - steel-man argument: presenting counter argument in its strongest form
 - don't straw man: misrepresenting opponent's argument

Property: Updating - Reflective & Frequent

- **Definition:** Treating predictions not as final pronouncements but as dynamic states reflecting current knowledge.
- **Why it Matters:**
 - allows for improvement
 - theoretical consistent (Bayes rule)

Property: Mindset - Growth Mindset

- **Definition:** Believing that prediction skill that can be developed through practice (Growth Mindset).
- **Why it Matters:**
 - Motivates continuous learning and effort.
 - Prevents discouragement from bad results

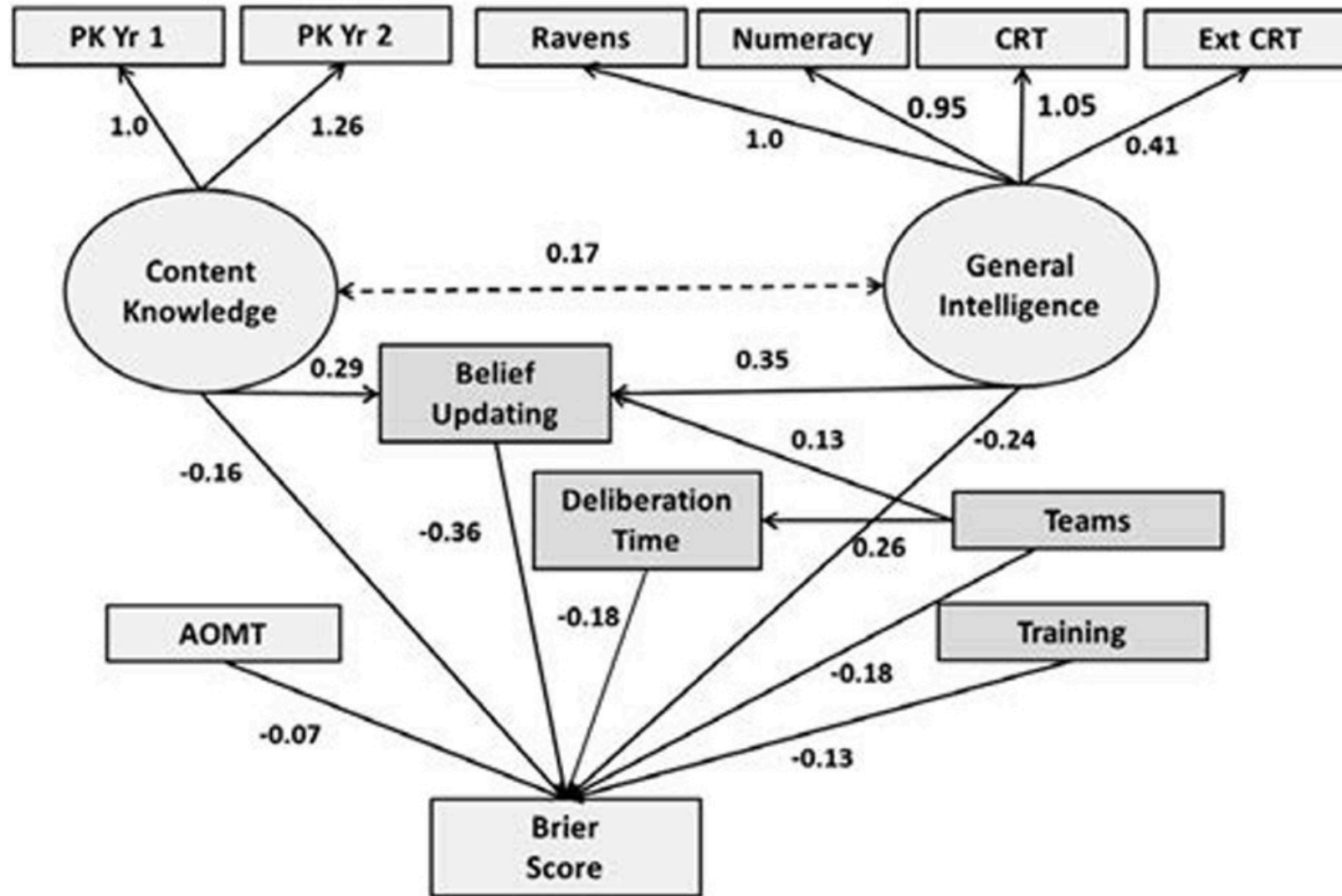
Add-on: Teams are superior

- teams of forecasters outperformed single forecasters
- discussion and social ties improved time investment and effectiveness

Quantitative estimates

- concepts are strongly interconnected
- difficult to measure the exact contribution of each property
- next slide shows result from (Mellers et al. 2015)
 - participants predicted political events
 - SEM (structural equation modeling)
 - structure of observed and unobserved variables
 - $A \rightarrow B$: A influences B
 - $A \leftrightarrow B$: A and B correlated
 - Directions chosen by researchers
 - AOMT: actively open-minded thinking.

Structural equation result



Black swans

- The **superforecaster** agenda argues
 - find precise prediction targets
 - follow good prediction habits
 - you can achieve **quantification of uncertainty**
- Now to “**black swans**”(Taleb 2007):
 - rare
 - real
 - relevant, but ignored (!)
- black swans are everywhere and unavoidable (Taleb 2007)
 - uncertainty quantification of superforecasters an illusion (Taleb et al. 2023)

Predictable or not?

- Clearly defined events are probabilistically predictable, but ...
 - real world decision making often does not depend on single event
 - calibration/performance becomes essentially unknowable for extremely unlikely events
 - adaptation to risk generates new risk (e.g., the financial system)
 - definition of appropriate events imperfect
 - world too unstable (technological progress, climate change)

Illustration: Extremely unlikely events

- Consider a prediction $\hat{p} = 0.001$
- we conjecture that the actual probability is $p = 0$
 - uncalibrated prediction
- we need to observe a lot of false event realizations to infer that kind of misscalibration

Illustration: Intractability of real-valued prediction targets

- for risk management we are often most interested in extreme events
 - heavy rain fall
 - strong tectonic movements for earthquakes
 - large losses of financial institutions
- if we do statistical inference based on iid samples
 - most observations are not in the tail
 - inference heavily relies on shape assumptions of the distribution
- if implications/utility/loss scales strongly with magnitude, expected loss (or its variance) cannot be inferred from a sample of data

Conclusion

- there are measures to improve predictability and appropriate reflection of uncertainty
 - probabilistic thinking
 - intellectual humility
 - active open-mindedness
 - frequent and deliberate updating
 - growth mindset
 - team work
- risk and uncertainty nowhere near to being defeated
 - in fact, may be impossible

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